Exp: 10 a) Internetworking using Router:

Aim:

To implement and design configure a simple internetwork using a router and two PC's in CISCO packet tracker and verify network connectivity.

Procedure:

Open CISCO Packet Tracer and create a new network topology.

Place One Router and 2 PC's in the workspace.

Connect PC0 and PC1 to the router using copper straight through cables.

Configure Router1 Interfaces with the following

Fast Ethernet 0/0 → 192.168.10.1/24

Fast Ethernet 0/1 → 192.168.20.1/24

Enable both router interfaces using the no shutdown command.

Configure PC0 with

IP Address → 192.168.10.2

subnet mask → 255.255.255.0

Default gateway → 192.168.10.1

Configure PC1 with

IP Address → 192.168.20.2

Subnet Mask → 255.255.255.0

Default gateway → 192.168.20.1

Send a simple PDU from PC0 to PC1

Verify that the PDU from reaches PC1 successfully & an acknowledgement is Received.

```
Router>en
Router#conf term
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#do show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  administratively down down
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

```
Router(config)#int gigabitEthernet0/0
Router(config-if)#ip address 10.0.0.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

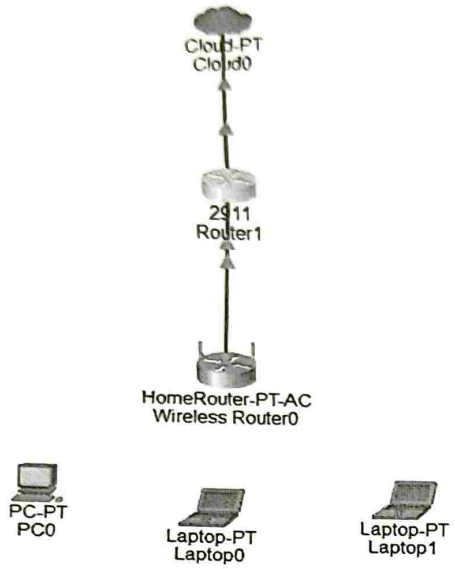
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int g0/1
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

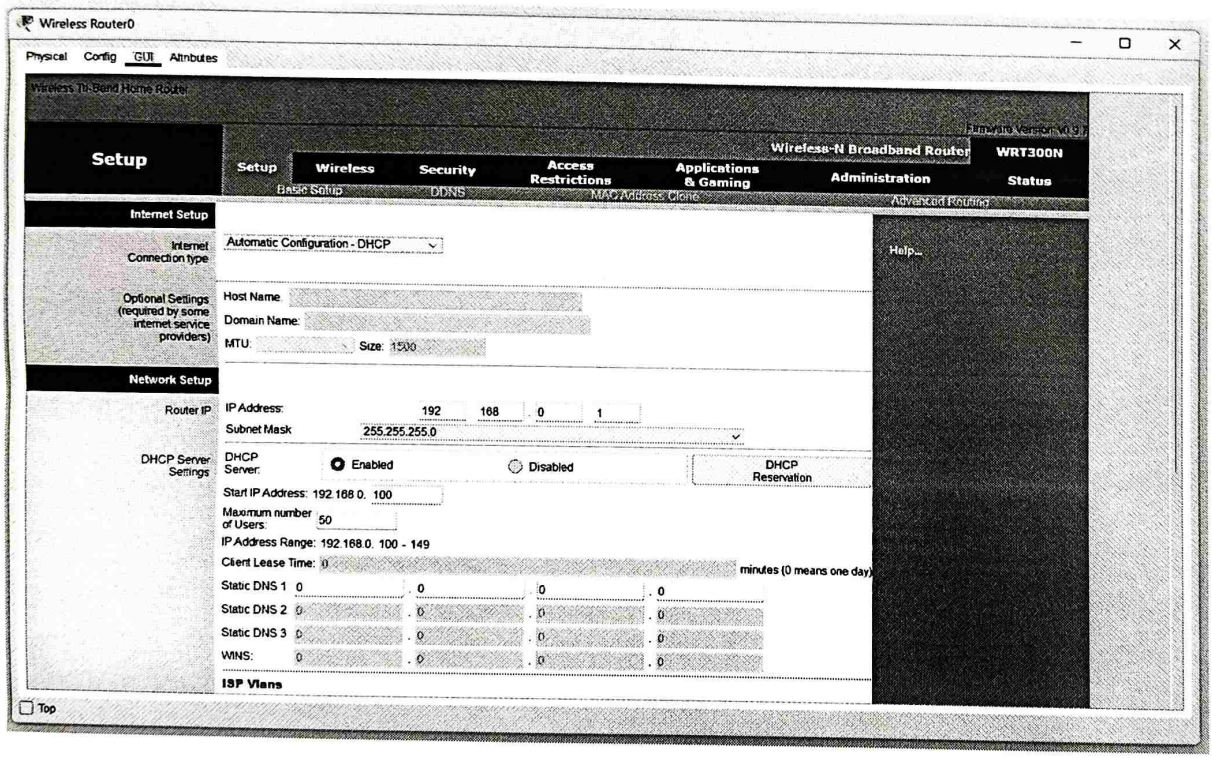
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#end
```



Home WIFI Router:



Result :

A simple network using a Router and ~~two~~ PC's was successfully designed & configure.